Sona College of Technology
Department of CSE

QKD-Based Secured Communication Framework for Sensitive Data Sharing Using Multimodal Biometric Authentication

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Team No: 4

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Project Guide
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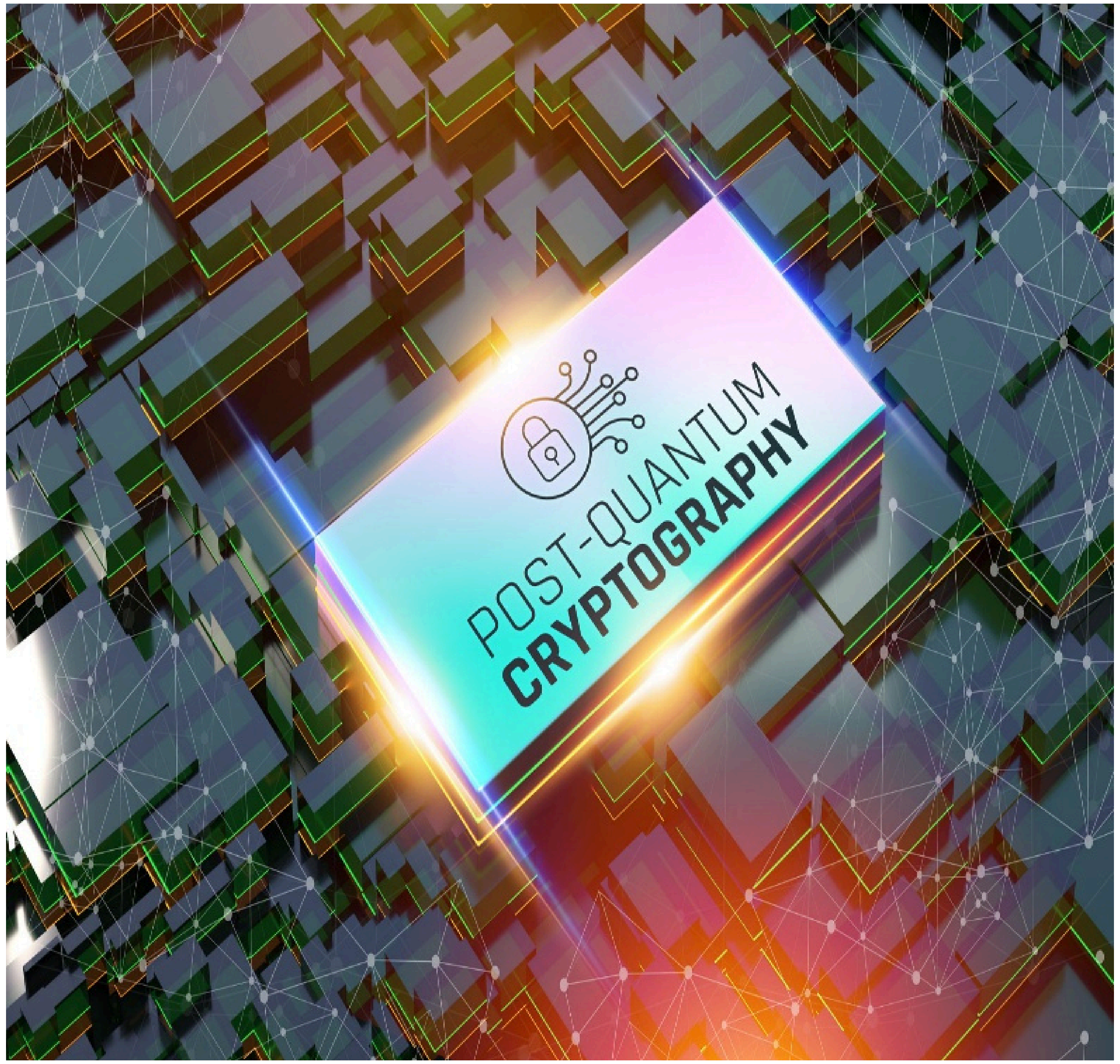
Problem Statement

- Quantum computing threatens current encryption standards.
- No eavesdropping detection in classical key exchange.
- Password token-based authentication is vulnerable.
- Lack of integration between quantum security and biometrics.
- Biometric key generation is inconsistent.
- Slow migration toward post-quantum cryptography.



Objective

- To ensure highly secure sharing of sensitive data
- To implement Quantum Key Distribution (QKD) for secure key generation
- To achieve information-theoretic security using AES-256 encryption
- To integrate multimodal biometric authentication
- To bind biometric identity with quantum-generated keys
- To prevent replay attacks and key misuse
- To enable real-time eavesdropping detection
- To prepare systems for the post-quantum security era

Literature Review

S. No	Author Name	Title	Methods/Techniques/ Algorithms	Advantages	Drawbacks / Limitations
1	C. H. Bennett, G. Brassard	Quantum Cryptography: Public Key Distribution and Coin Tossing	BB84 Quantum Key Distribution protocol	Provides unconditional security and detects eavesdropping	Requires specialized quantum channel
2	V. Scarani et al.	The Security of Practical Quantum Key Distribution	QKD security analysis, QBER detection	Strong theoretical and practical security validation	Performance affected by noise and channel loss
3	A. K. Jain, A. Ross	Multimodal Biometrics for User Authentication	Fingerprint and face biometric fusion	Improved accuracy and reduced spoofing	Higher system complexity
4	S. Sharma, A. K. Singh	Survey on Biometric Cryptosystems	Biometric key binding, hashing techniques	Enhances user-centric security	Sensitive to biometric quality
5	H. E. Mozo et al.	Quantum-Classical Hybrid Encryption Framework	BB84 simulation with classical encryption	Practical integration of QKD in real systems	Uses computational security for data encryption

Literature Review

S. No	Author Name	Title	Methods/Techniques/ Algorithms	Advantages	Drawbacks / Limitations
6	A. Raj, V. Balachandran	Hybrid Encryption Using QKD and Post-Quantum Cryptography	QKD + post-quantum algorithms	Resistant to quantum attacks	Increased computational overhead
7	S. Arumugam et al.	Hybrid Encryption Using Biometric Key	Biometric entropy-based key generation	Eliminates password dependency	Biometric variation affects consistency
8	K. Gkouliaras et al.	NuQKD: A Modular QKD Simulation Framework	BB84 simulation framework	Easy experimentation and scalability	Does not use real quantum hardware
9	L. Garms et al.	Integration of QKD with Post-Quantum Systems	QKD with secure key management	High security and future readiness	Deployment complexity
10	LL Scientific	Enhanced BB84 Protocol for Secure Communication	Improved BB84 with error correction	Better resistance to noise and attacks	More complex protocol design

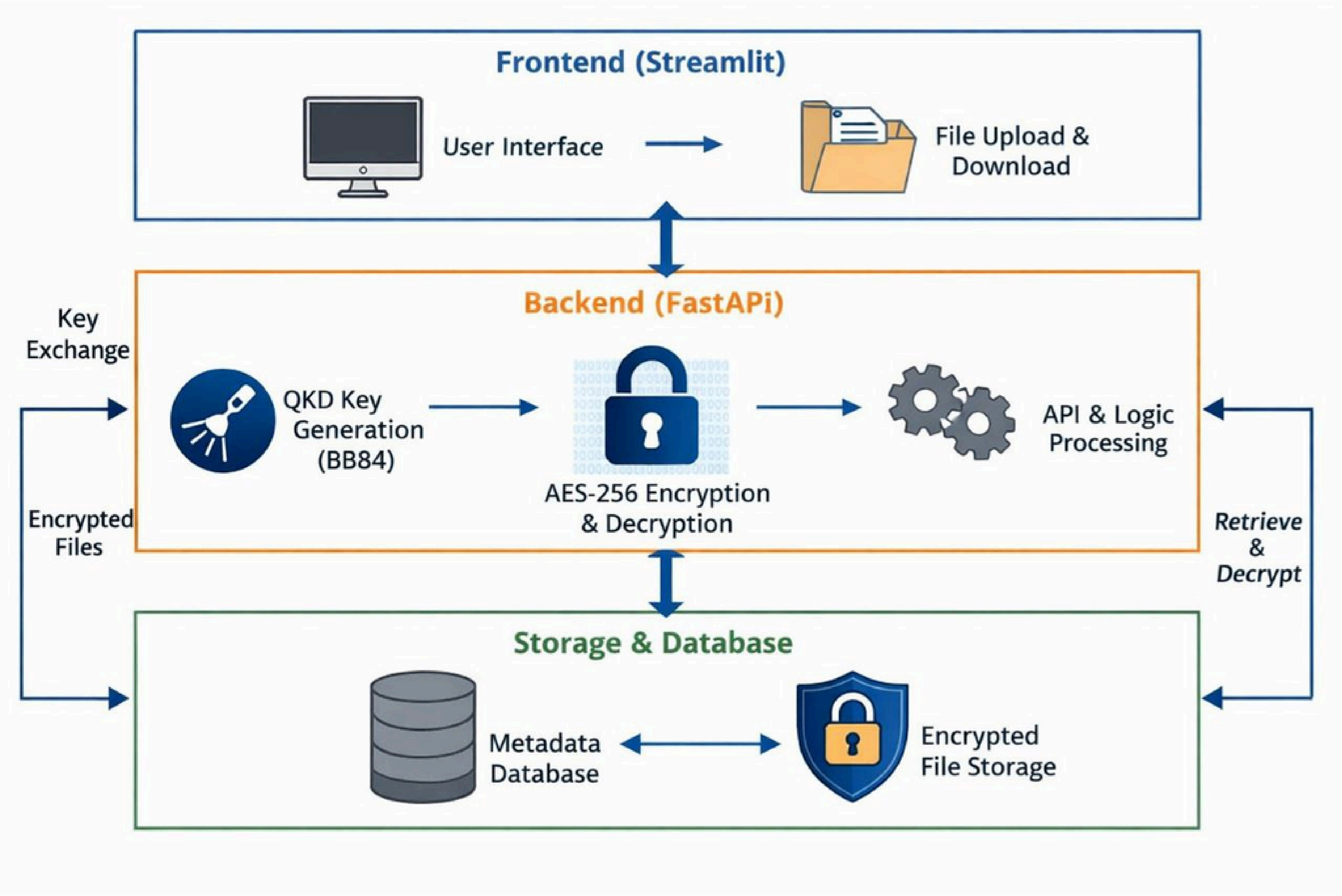
Gap Analysis

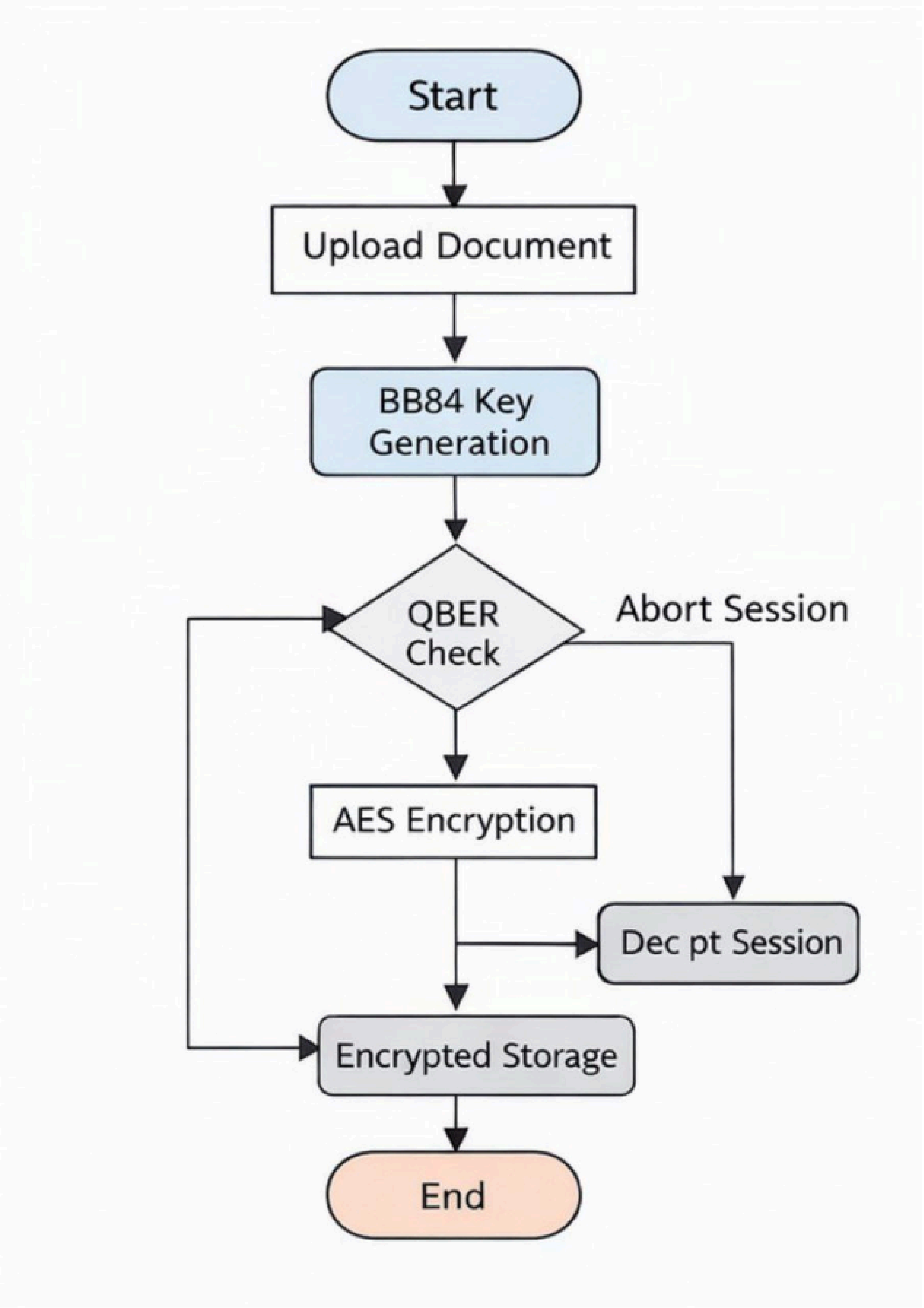
- Traditional encryption algorithms are vulnerable to future quantum computing attacks.
- Classical key exchange lacks real-time eavesdropping detection mechanisms.
- Single biometric authentication is prone to spoofing and accuracy limitations.
- Biometric key generation produces inconsistent results across captures.
- Hardware-based QKD solutions are costly and difficult to deploy widely.
- Password-based authentication introduces phishing and credential theft risks.

Gap Analysis

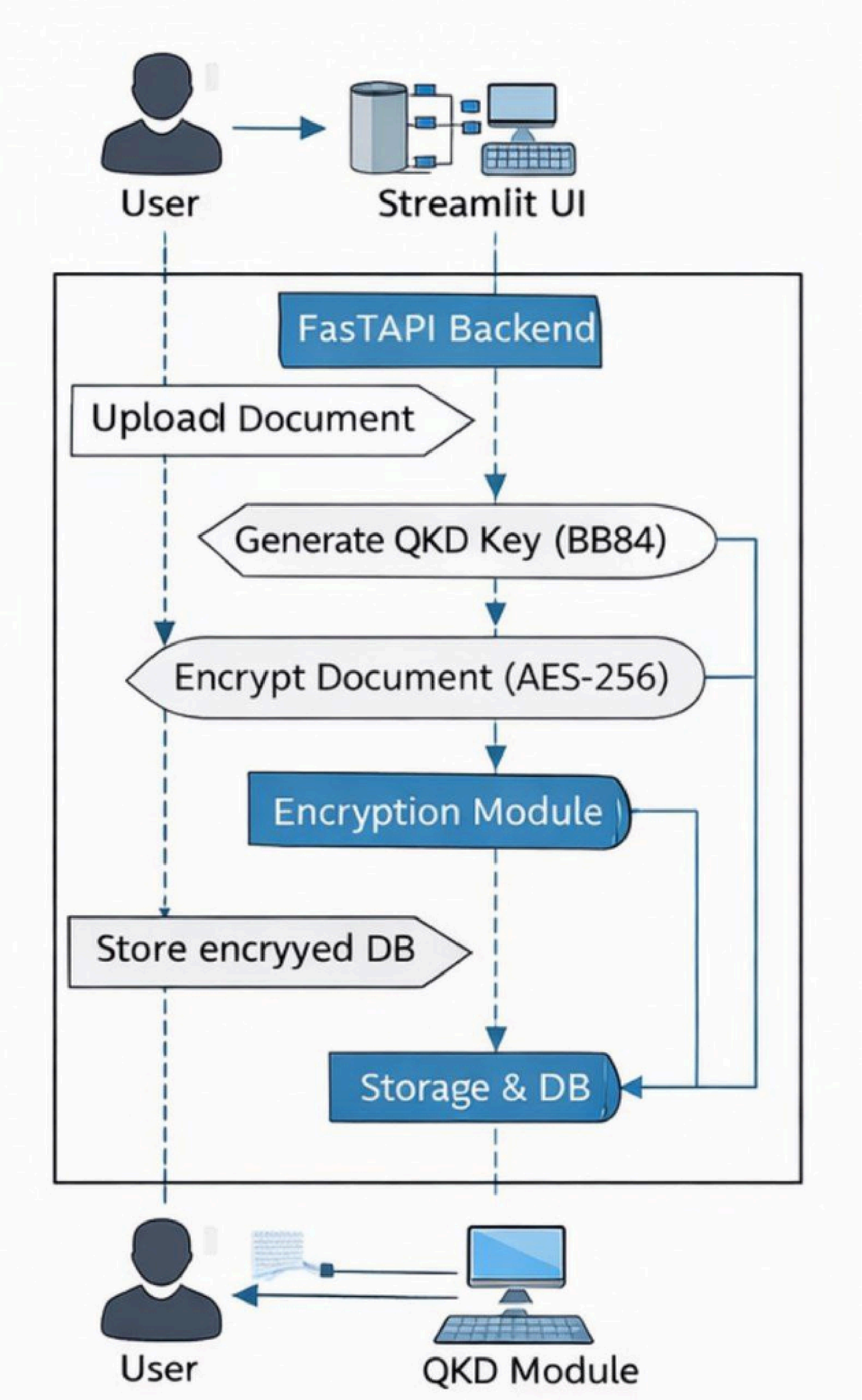
- Session keys stored on disk increase the risk of key compromise.
- Cloud-dependent encryption exposes data to third-party access risks.
- Lack of standardized key regeneration mechanisms causes decryption failures.
- Existing biometric enrollment systems are complex and reduce usability.
- High authentication latency affects user experience and workflow efficiency.

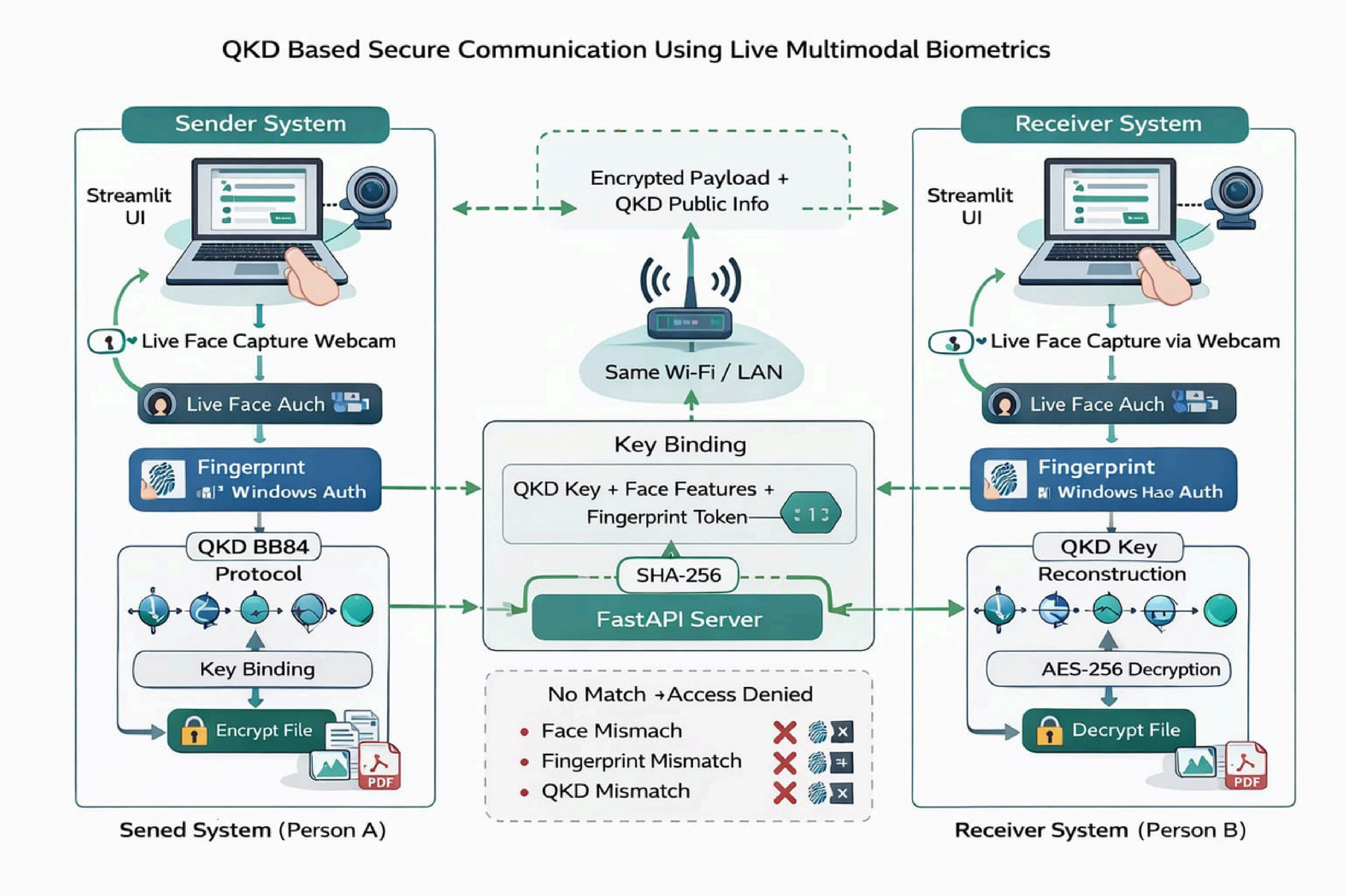
System Architecture/Design





System Architecture/Design





Methodology/Implementation

1.Technological Stack

- **Backend Framework:** FastAPI (Python)
- **Frontend:** Streamlit
- **Cryptography:** Python cryptography library (Hazmat)
- **Face Recognition:** DeepFace + OpenCV
- **Fingerprint Module:** pyfingerprint + serial communication
- **QKD Simulation:** NumPy + secrets module
- **Key Derivation:** HKDF-SHA256

2. Module Structure

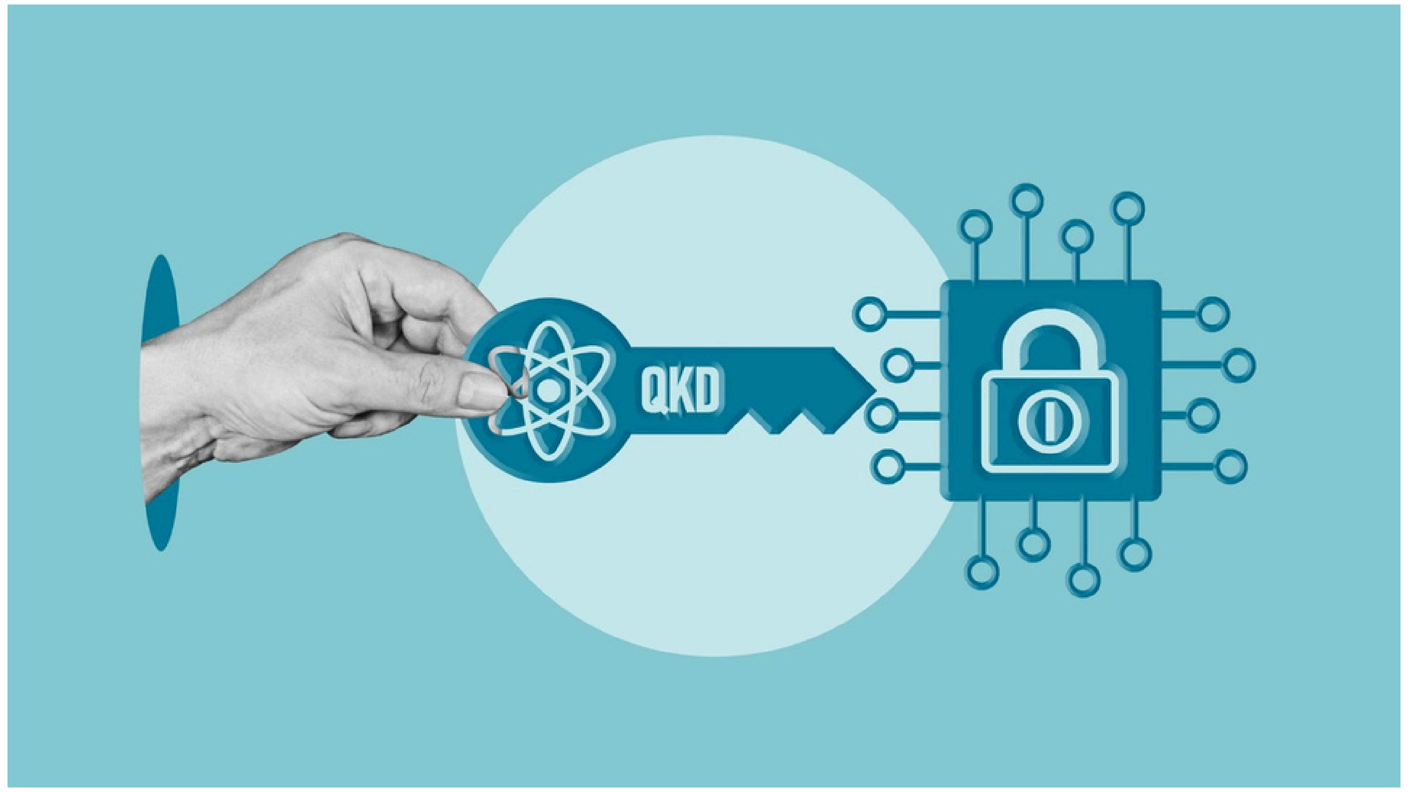
Core Modules:

- bb84.py — Quantum key generation & eavesdropping detection
- face_auth.py — Face enrollment, embedding extraction, verification
- fingerprint_auth.py — Fingerprint capture & template matching
- key_fusion.py — Multi-factor key derivation using HKDF
- aes_crypto.py — File/byte encryption with AES-256-GCM
- security_validators.py — Input sanitization & file validation

Methodology/Implementation

Application Layer:
main.py — FastAPI backend with REST endpoints
streamlit_app.py — Interactive web interface
config.py — Centralized configuration & paths

Enrollment Scripts:
enroll_face_sender.py / enroll_face_receiver.py
enroll_fingerprint_sender.py / enroll_fingerprint_receiver.py



3. Workflow Implementation

Sender Side:

- Run enrollment scripts (one-time face + fingerprint registration)
- Launch Streamlit on port 8501
- Authenticate via live face capture + fingerprint scan
- System generates BB84 key + fuses with biometric hashes
- Upload file → Compress → Encrypt with fused AES key
- Download .enc file for transmission

Receiver Side:

- Complete enrollment process (separate templates)
- Launch Streamlit on port 8502
- Authenticate with live biometrics
- System retrieves sender's QKD key from shared state
- Reconstructs identical AES key via same fusion parameters
- Upload .enc file → Decrypt → Download original

Methodology/Implementation

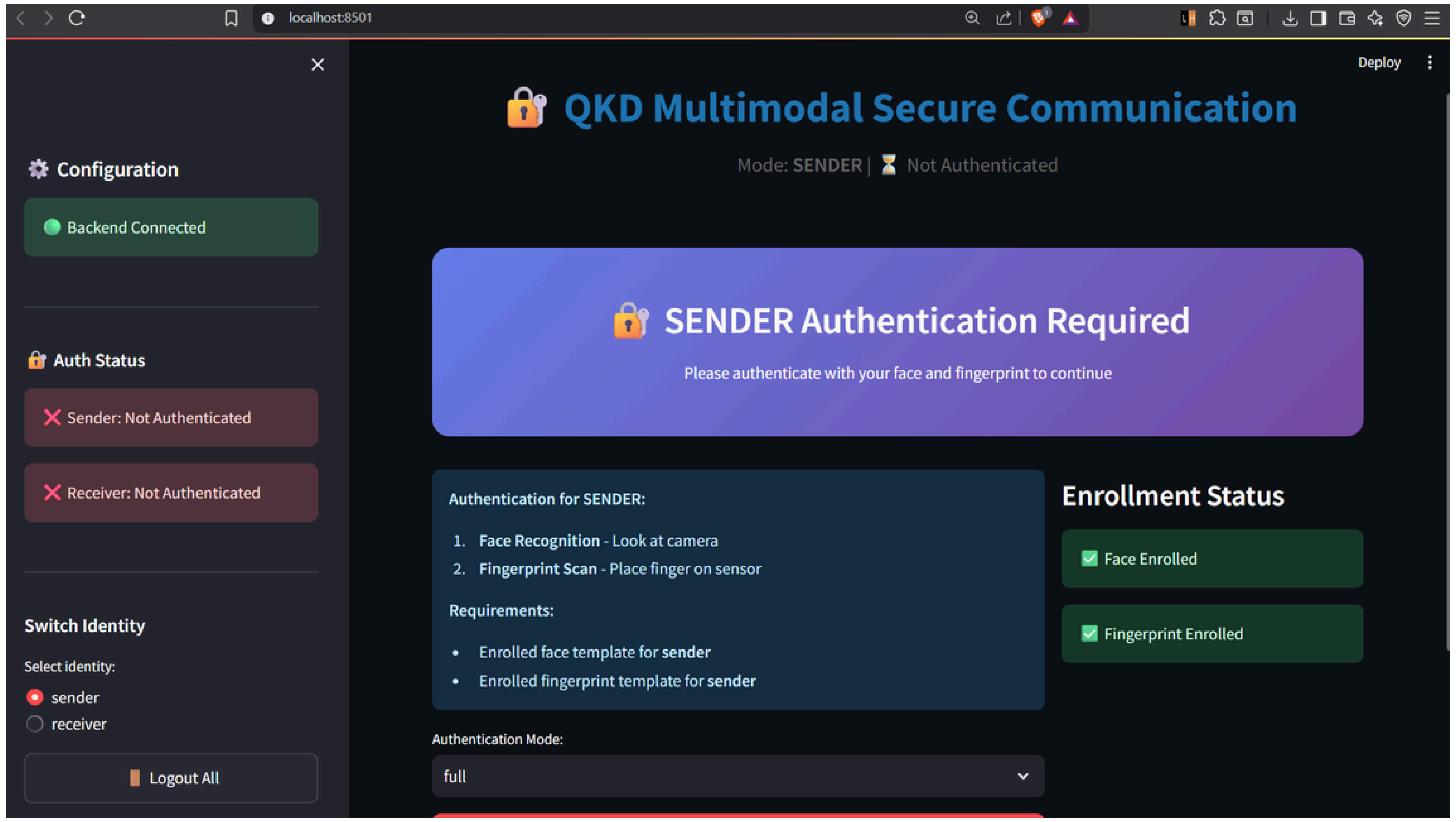
4.API Endpoints (Point-wise)

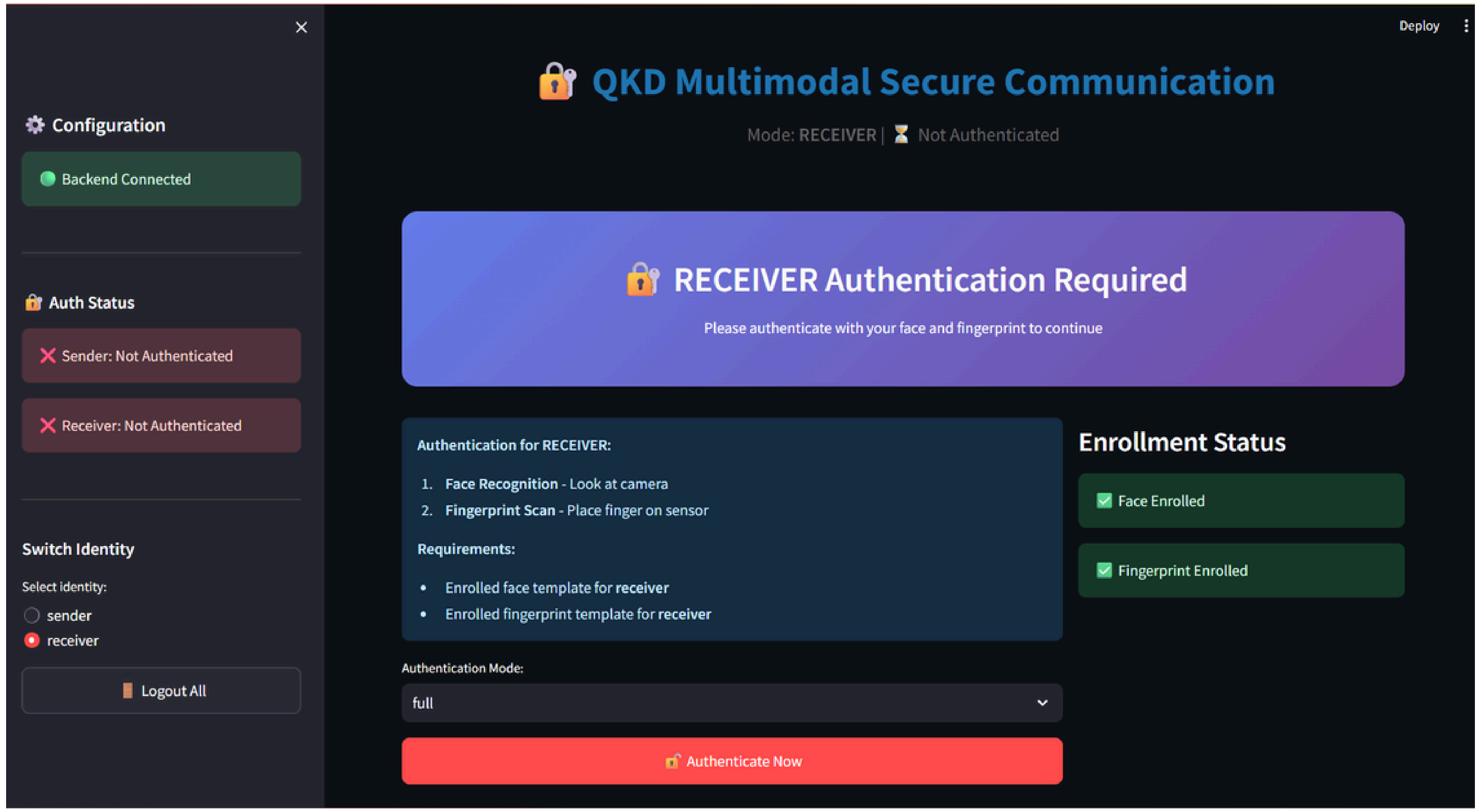
- **/authenticate** — POST — Performs biometric verification and key generation.
- **/switch_identity** — POST — Switches user role between sender and receiver.
- **/encrypt** — POST — Encrypts files (sender side only).
- **/decrypt** — POST — Decrypts files (receiver side only).
- **/qkd/generate** — POST — Generates standalone QKD keys.
- **/enrollment/{id}** — GET — Checks biometric enrollment status.
- **/session/{id}** — GET / DELETE — Handles session management operations.

5. Security Features Implemented

- **Eavesdropping Detection**: BB84 error rate monitoring (threshold: 15%)
- **Multi-Factor Binding**: Key tied to QKD + face + fingerprint
- **No Key Storage**: Ephemeral session keys (memory-only)
- **File Validation**: MIME-type checking + extension whitelisting
- **Input Sanitization**: Path traversal protection, size limits
- **Authenticated Encryption**: AES-GCM provides confidentiality + integrity

Demo/Screenshots





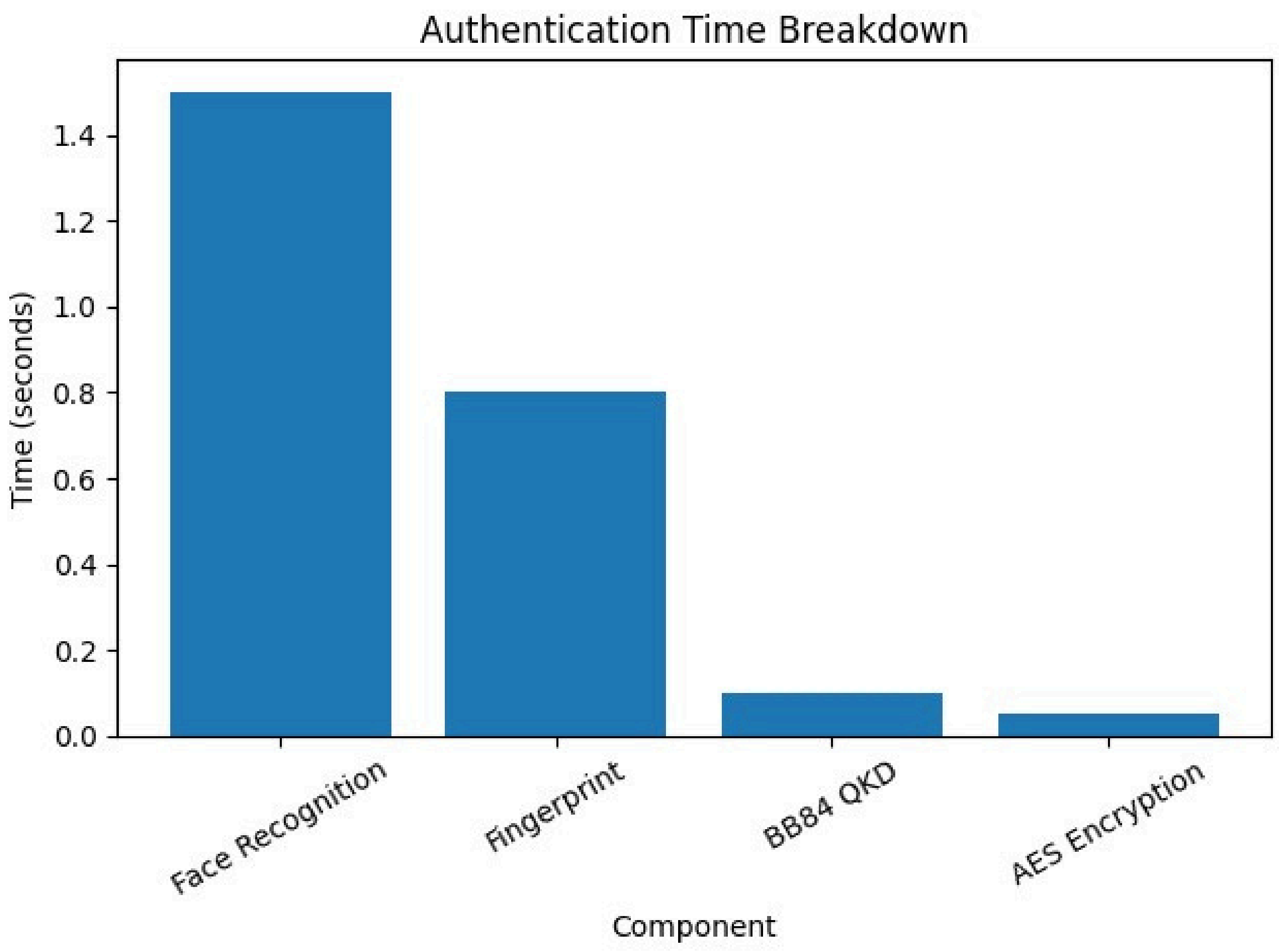
Results

1. Authentication Performance

- Face recognition accuracy: ~95% with ArcFace model
- Authentication time: 3–5 seconds total
- Face capture: 1–2 seconds
- Fingerprint scan: 0.5–1 second

2. Encryption Performance

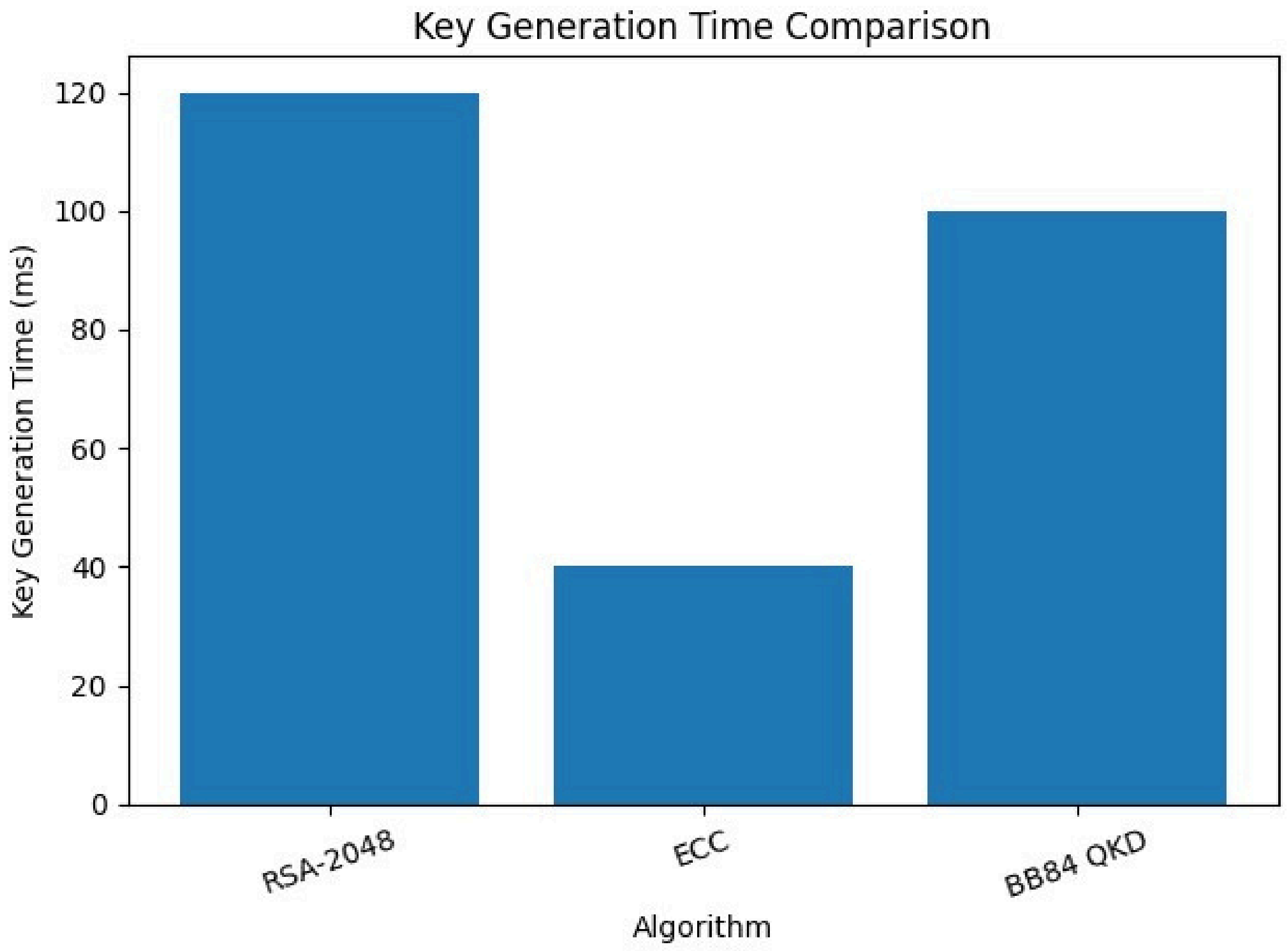
- BB84 key generation: ~100ms
- File encryption (1MB): ~50ms
- AES-256-GCM throughput: Optimized for real-time use



Results

3. Security Metrics

- Eavesdropping detection: >15% error rate triggers alert
- Key entropy: 256-bit fused key (QKD + biometrics)
- Zero key reuse: Session-based ephemeral keys
- Multi-factor binding: 3-factor authentication (QKD + face + fingerprint)



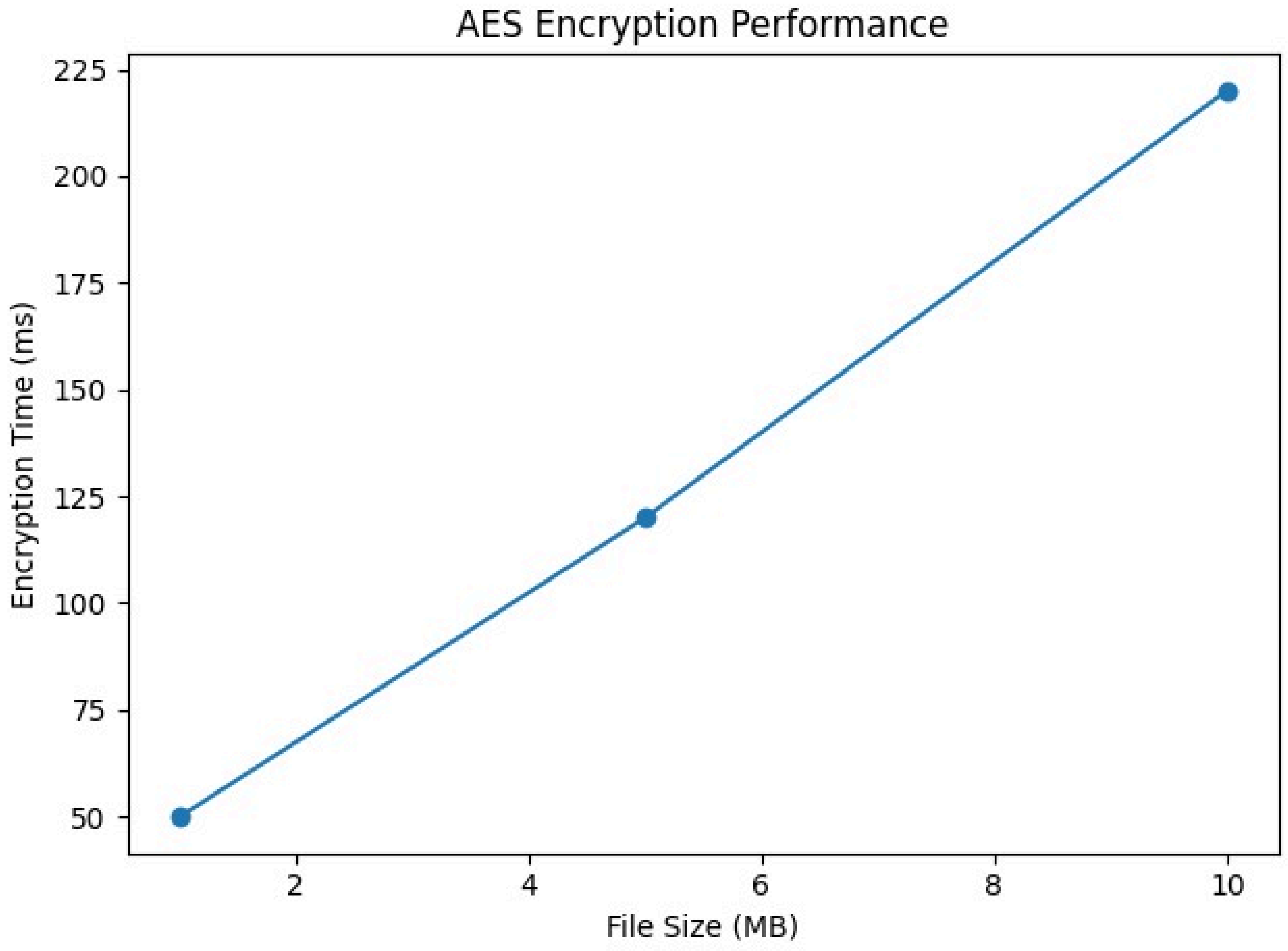
4. System Validation

- Successful sender-receiver file transfer
- Cross-identity key synchronization verified
- Decryption only possible with authenticated receiver biometrics
- Session isolation: No key leakage between sessions

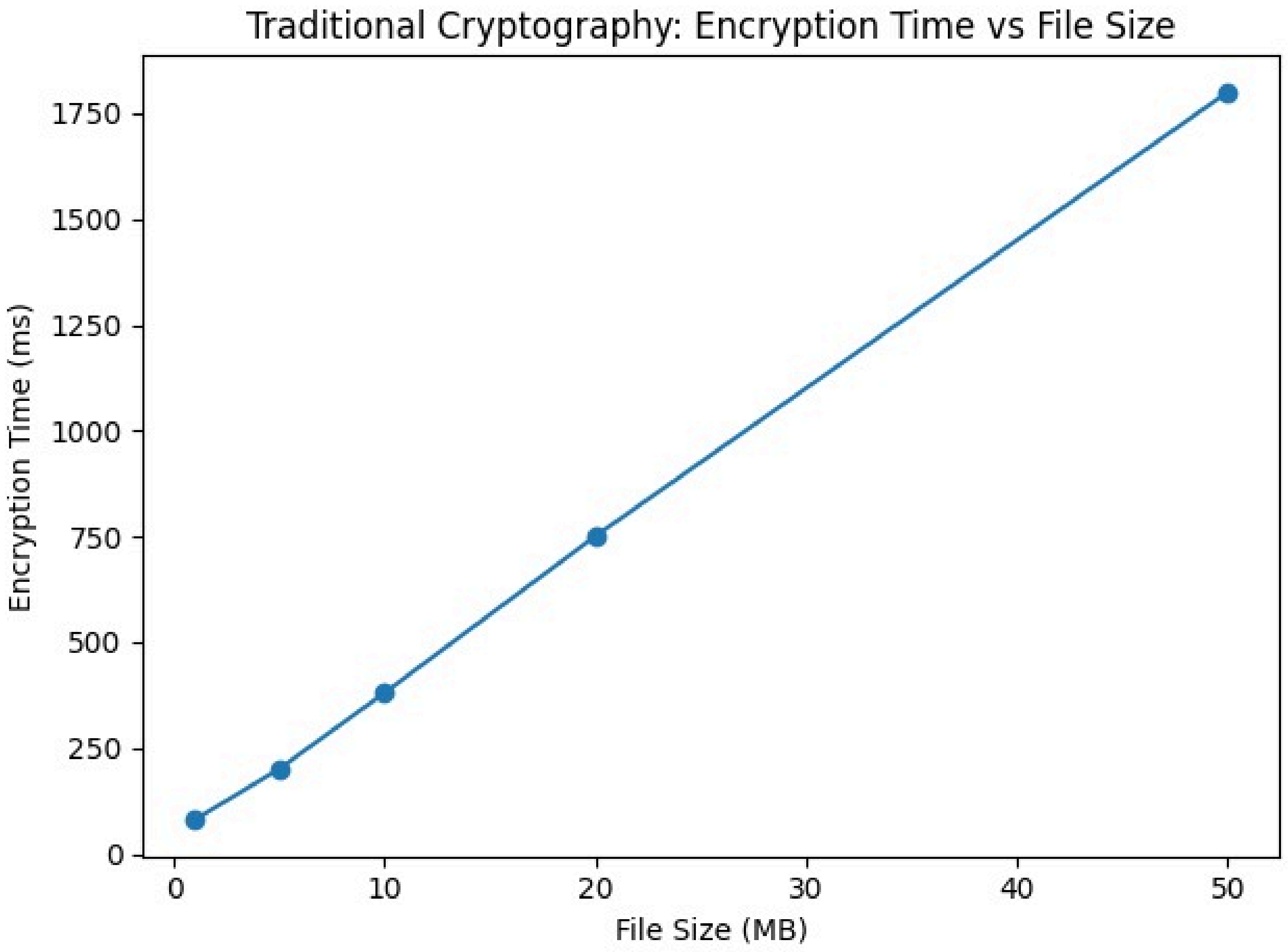
Results

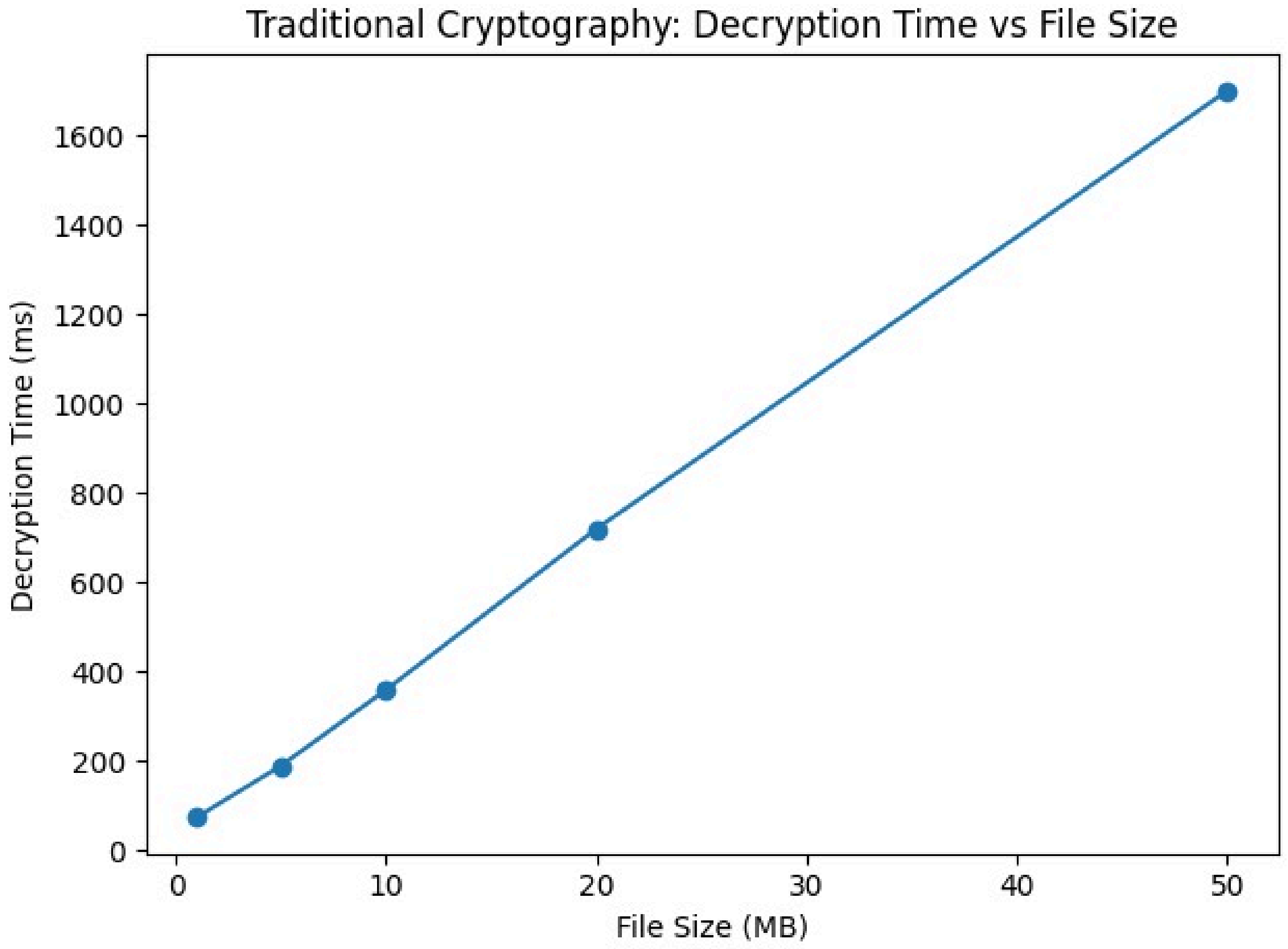
5. Demo Capabilities

- Live face capture with OpenCV
- Hardware fingerprint sensor auto-detection
- Real-time encryption/decryption workflow
- Web-based UI with role switching

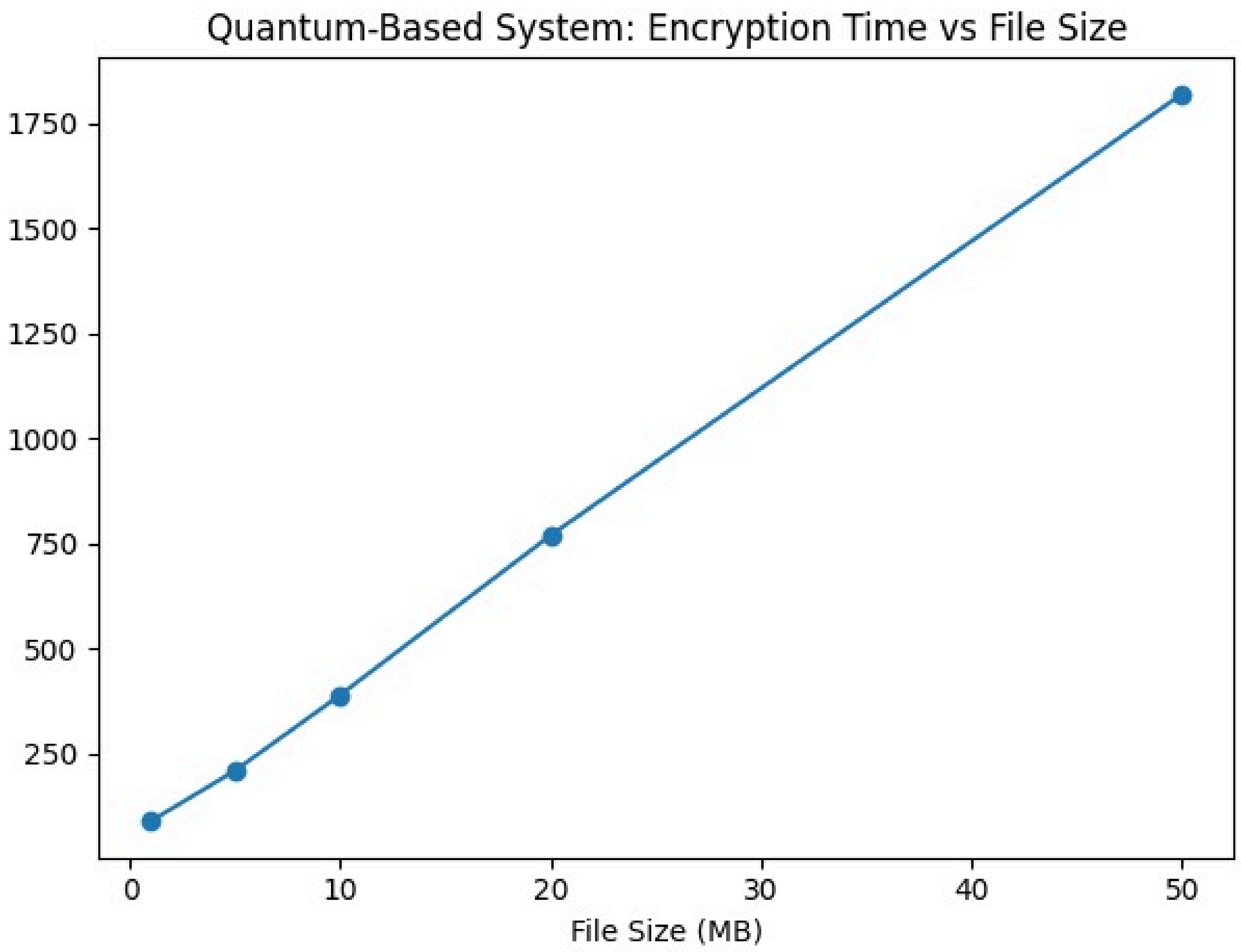


Graph

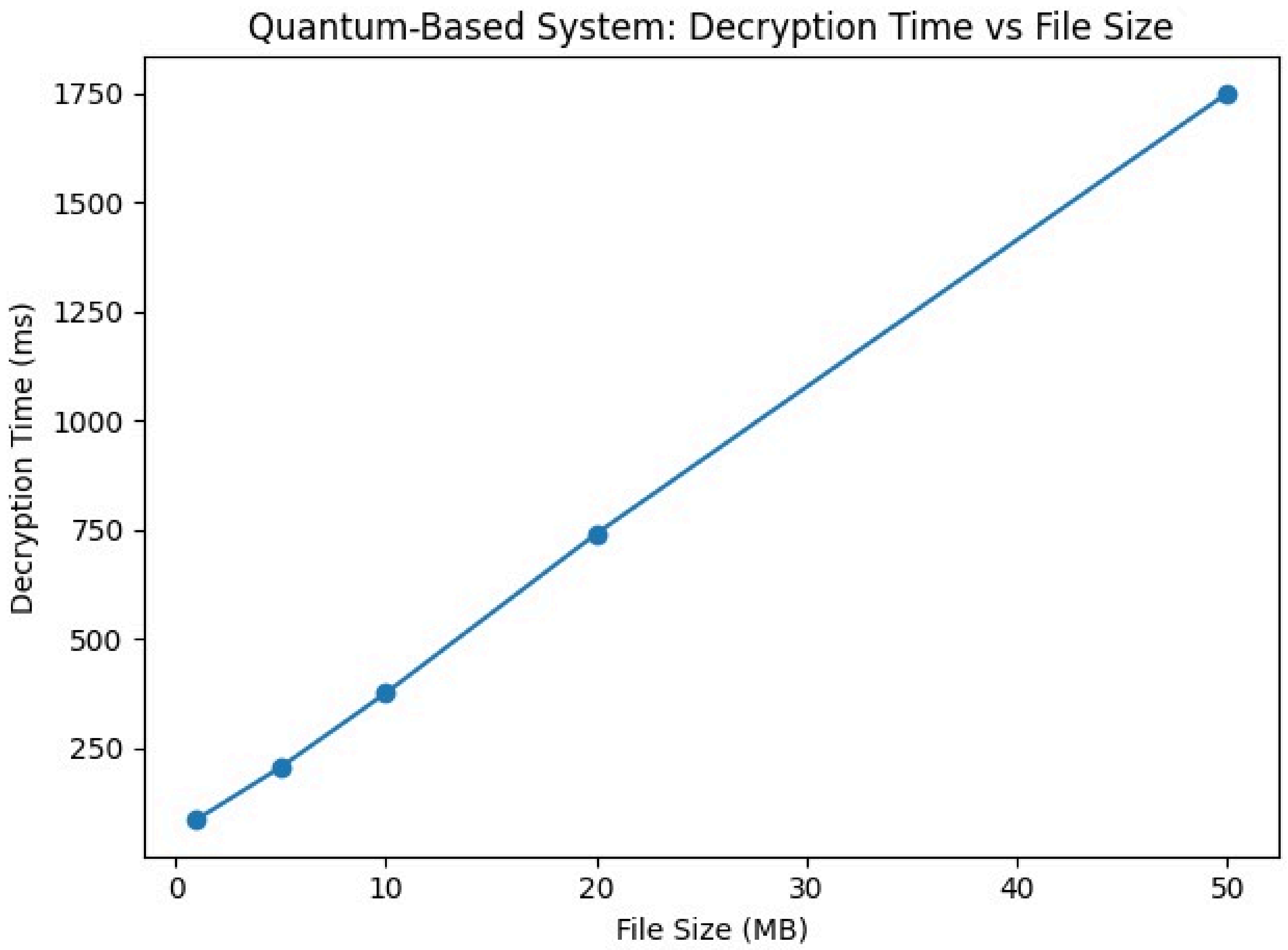

Traditional Cryptography: Encryption Time vs File Size
Encryption Time (ms)
File Size (MB)
0
250
500
750
1000
1250
1500
1750
0
10
20
30
40
50


Traditional Cryptography: Decryption Time vs File Size
Decryption Time (ms)
File Size (MB)
0
200
400
600
800
1000
1200
1400
1600
0
10
20
30
40
50

Graph



Quantum-Based System: Encryption Time vs File Size
1750
1500
1250
1000
750
500
250
0
10
20
30
40
50
File Size (MB)



Quantum-Based System: Decryption Time vs File Size
Decryption Time (ms)
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File Size (MB)

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